Can Consistent Forces Recover Molecular Shapes?

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Abstract

A molecule is a group of atoms, the tiny pieces of matter. Its shape is their arrangement in three dimensions. Forces push and pull those atoms; a model learns force predictions from examples. Energy is a number associated with the arrangement. Energy-derived models calculate forces from negative energy slopes: how energy changes with atom movement. Direct models predict force arrows; consistency training penalizes disagreement with the energy-derived arrows.

We compare these approaches at similar force error and computing cost on aspirin. Direct and consistency models meet our matching rule at one consistency strength. This rule uses validation examples reserved for model selection. All five tested models, including the energy-derived comparison model, recover zero of ten targets. All targets appear in training, but this pilot cannot rank recovery performance. Inaccurate forces at reference shapes and excessive coordinate steps leave the recovery question open.

1 Why this problem matters

Repeated physics calculations can make molecular simulation expensive. A learned model offers a faster approximation. Training changes its adjustable numbers, called parameters, to reduce prediction errors against reference examples.

Energy is a number associated with an atomic arrangement. A stretched spring gives a simple analogy: changing its shape changes stored energy and the pull it produces. Energy slope means how quickly energy changes when an atom moves. An energy-derived model predicts energy, then calculates force arrows from the negative slopes. A direct-force model predicts the arrows themselves. Our consistency training adds a penalty when these arrows disagree with forces calculated from negative energy slopes.

Our objective is to compare correct-shape recovery at similar force error and measured computing cost. Recovery means that the search reaches its assigned target shape and meets the force stopping rule.

2 What earlier studies establish

Landscape17 supplies molecular shapes, reference forces and paths between stable arrangements [1]. It motivates checking final shapes, beyond prediction errors. Bigi et al. describe problems when direct forces do not follow one energy function [2]. Their findings motivate separating a force-based stop from structural success.

We build our models with E3x, software for networks that obey three-dimensional symmetry rules [3]. MACE supplies a separate established model for predicting energies and forces [4]. Earlier work already studies physical constraints in training losses [5]. We compare error, cost and recovery under shared conditions; consistency training itself is not new.

3 Data and a fair comparison

We use aspirin from Landscape17 [1]. Each molecule has 21 atoms. Reference energies and forces come from hybrid density-functional theory, a detailed calculation based on electron physics. These reference labels are calculated answers, not exact experimental truth.

A transition state is a high-energy arrangement with two downhill branches toward stable arrangements, called minima. A connection group contains the records from both branches of one transition state. Both branches stay in the same partition. Training adjusts parameters; validation selects settings and saved models; the test partition is reserved for later trained-model evaluation.

Training/validation/test contain 26/5/6 connections and 4,891/974/1,127 records. Training-only normalization puts energy and force errors on comparable scales. Some groups share minima, so records are not independent examples of new destinations. Empirical branch classifications corrected 26 of 74 targets previously assigned from filename ordering. Earlier model-free software checks parsed test coordinates and labels. No trained test scoring has occurred.

3.1 Inputs, outputs and the three model types

Our E3x models are graph neural networks. Atoms are nodes, and nearby atom pairs form links. Successive calculation stages, called layers, pass learned information along these links. Inputs are atom types and three-dimensional positions; outputs are one molecular energy and a force arrow for each atom. All arms use 64 feature channels, containing learned numbers and vectors. Two interactions pass messages; eight radial functions describe pair distances.

Links extend to 5 Å, where an angstrom (Å) is 10^{-10} metres. Internal features carry both scalar numbers and vectors with direction. Rotating a molecule rotates its predicted forces without changing its energy. This property is called equivariance for forces and invariance for energy. The energy model has 43,521 parameters; models with separate energy and force outputs have 43,649.

Our baseline, the energy comparison model, calculates $\widehat{\mathbf{F}}^E = -\nabla_{\mathbf{R}}\widehat{E}$. Here, $\widehat{E}$ is predicted energy, $\mathbf{R}$ contains atom positions, and the gradient gives energy slopes along position coordinates. Both direct arms also predict energy trained against reference labels. Ordinary direct training has no agreement penalty; consistency training adds one. They are not force-only models.

3.2 Training loss and measured success

The loss is the number that training tries to reduce. Our loss adds normalized squared energy error, normalized squared force error and, where applicable, force–energy disagreement:

$$\mathcal{L} = w_E \,\mathrm{mean}(e_E^2) + w_F \,\mathrm{mean}(e_F^2) + \lambda \,\mathrm{mean}\left[\left((\widehat{\mathbf{F}}^D - \widehat{\mathbf{F}}^E)/s_F\right)^2\right]. \tag{1}$$

Here, e_E and e_F are prediction errors divided by their training-derived scales. The force scale is s_F. The superscript D means directly predicted force. Weights $w_E = 0.014584365670390719$ and $w_F = 1.9854156343296092$ balance supervised errors. The final term is absent in the energy baseline. Direct training uses $\lambda = 0$; consistency training uses a positive λ.

Final E3x fits use Adam to update parameters, learning rate 0.003 and 64-bit floating-point arithmetic. Each update samples four records; records can repeat. Each fit has 200,000 updates: 800,000 sampled exposures. Two seeds, specifying random initialization and sampling, partly check repeatability.

Validation occurs every 500 updates, including zero. BEST is the first saved model state, or checkpoint, with minimum validation force error.

Our main score is force mean absolute error (MAE). We average mistakes without their signs across the three coordinate directions of each force arrow. We then give each validation connection equal weight. Smaller values are better.

Scores use meV/Å: a millielectronvolt is one thousandth of an electronvolt, an energy unit. Energy divided by distance measures force.

Recovery additionally requires maximum predicted force component ≤ 0.01 eV/Å and aligned structural RMSD ≤ 0.05 Å. Root-mean-square deviation (RMSD) measures typical atom-position disagreement after rotation, translation and allowed atom-symmetry matching. Small predicted force is a stopping condition; it is not small prediction error.

4 What we trained and learned

4.1 A fitting plateau changed the training recipe

Early small-subset checks plateaued. MACE Run B fitted the same 128 frames, reducing force squared error to 0.001515 of predicting zero. Different settings prevent a controlled comparison. The plateau did not establish an unavoidable data or energy-derived-force limit.

Later 1,000-update energy and direct fits selected update zero as BEST: training improved energy but worsened force predictions.

We then completed three 200,000-update energy baselines (Figure 1). A used width 32 and learning rate 0.001; B used width 64 and 0.003. Both used equal normalized energy/force weights. C retained B's structure and rate but used the balanced weights in Section 3.2. Width counts internal feature channels.

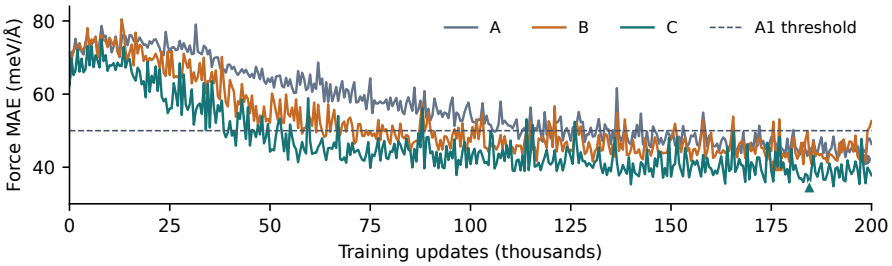


Figure 1: Our energy-baseline validation results. Lower force MAE is better. Markers show BEST; the dashed line is the declared 50 meV/Å adequacy rule. Each point uses all 974 validation records, with equal connection weighting.

C reached 34.587 meV/Å, versus 40.370 for B and 42.108 for A. It passed the adequacy rule. Predicting zero force gives 62.504 meV/Å, so C reduces that error by 44.664%. However, C's energy error is higher than B's: 126.990 versus 28.178 meV per molecule. We selected C on the energy model and transferred its recipe without retuning the other model types.

4.2 Agreement improves, but recovery still needs a separate test

We reused C seed zero and trained nine more fits, giving two seeds per row in Table 1. Consistency strength 0.1 reduces normalized force–energy disagreement squared error to 0.49–0.61, versus 116–379 for direct training. Its measured fit-plus-score time is 1.985 times direct prediction, excluding other diagnostics and overhead. This is not inference speed.

Table 1: Validation error at BEST. Force columns use meV/Å; energy uses meV per molecule. Seed limits show two observed values, not confidence bounds.

Model	Mean force MAE	Seed min–max	Mean energy MAE
Energy-derived	34.893	34.587–35.198	128.725
Direct	38.186	37.291–39.082	76.684
Consistency $\lambda = 0.1$	38.024	36.773–39.275	87.006
Consistency $\lambda = 1$	34.683	33.723–35.643	112.341
Consistency $\lambda = 10$	39.725	39.060–40.390	161.662

Only strength 0.1 meets our declared matching rule against direct prediction. The mean-error gap must be within 10% of the smaller mean and within both seed spans. A span is maximum minus minimum.

BEST is optimistic because it selects among 401 checks. Late baseline errors vary by 2.2–3.1 meV/Å standard deviation; checks are correlated. Nine subsequent fits finish 0.1–7.4 meV/Å above BEST. The 0.162 meV/Å matching gap is smaller than this variation; matching does not establish equivalence.

MACE Run A independently scores 5.349 meV/Å with equal connection weighting after 2,048 epochs, or training-data passes. Architecture, exposure and hardware differ; this is an external reference, not a controlled efficiency comparison. MACE recovery remains unmeasured.

5 The shape-search pilot fails for every evaluated checkpoint

A relaxation procedure repeatedly moves atoms along predicted forces. We tested five saved E3x checkpoints from the same ten branch starting arrangements across five validation connections. Each branch has its own target; the six distinct target minima all appear in training. Leaving the starts unchanged recovers zero targets. Every evaluated checkpoint also recovers 0/10 (Table 2). This fails our recovery pilot gate; successful recovery cannot rank the methods.

The procedure proposes $\mathbf{R}_{\mathrm{trial}} = \mathbf{R} + \eta\widehat{\mathbf{F}}$, where η converts predicted forces into coordinate changes. It starts at 0.2 Å^2/eV and can grow to 1.0. Backtracking halves a failed trial's scale; each branch permits 1,500 force attempts.

The procedure checks force convergence first. Otherwise, forces pointing broadly in compatible directions can permit acceptance. There is no energy-decrease test and no per-atom movement cap. A scale limit does not cap movement when forces are large.

Table 2: All checkpoints recover 0/10 original targets. "Wrong stop" meets the predicted-force rule but misses the target shape. Reference-start controls begin at the six target minima; they are separate follow-up checks, called post-hoc analyses.

Checkpoint	Wrong stop	No convergence	Reference-start recovery
Energy C, seed 0	2	8	4/6
Direct, seed 0	4	6	1/6
Direct, seed 1	4	6	0/6
Consistency 0.1, seed 0	6	4	0/6
Consistency 0.1, seed 1	9	1	0/6

Across 50 model/branch outcomes, 25 stop with wrong geometry and 25 fail to converge. Reusing the same five connections does not create 50 independent paths. The reference-start controls favour the energy checkpoint under this procedure, but do not isolate a consistency-penalty effect.

5.1 Why a small predicted force can be misleading

A 155-evaluation check reproduced all 50 stopping flags. Every model exceeds the stopping force threshold at all 11 exact reference minima. The reference labels themselves stay below 2.745 meV/Å. This mismatch at exact reference shapes does not exclude nearby shapes meeting both tolerances.

All 23 force-converged direct/consistency endpoints have disconnected 5 Å neighbour graphs. Atoms in different fragments are beyond the model's interaction cutoff. These are not intact molecular shapes; internal fragment forces can remain. Only seven endpoints have every atom isolated; the direct model then predicts exactly zero force automatically. This is not evidence of a correct molecular minimum.

Four selected trace replays reproduced the saved endpoints. Two separated, with largest accepted atom movements of 436.721 Å for direct prediction and 268.047 Å for energy-derived forces. The other two remained connected, with maxima 0.086492 Å for consistency and 0.006083 Å for energy-derived forces. Both force-converged energy endpoints remain connected.

These observations expose model and procedure limitations, without proving sole procedure causality or real molecular break-up. No evaluated E3x model recovers an original target under this procedure. Better agreement on stored examples does not establish agreement everywhere or guarantee correct shapes. Zero recovery cannot establish method equivalence.

Evidence covers aspirin, five validation connections and two seeds, with shared minima. Validation selects settings and BEST; test generalization is unestablished. Failed and incomplete trials remain recorded.

6 A related representation check and the remaining work

Faculty feedback prompted four new scalar-only fits and reuse of two completed scalar-and-vector E64 fits. Scalars carry numbers; vectors also carry direction. All models still predict forces that rotate correctly with the molecule. E64 had lower validation error under its own fixed training recipe in both seeds. This preliminary observation does not isolate a feature-type effect or establish recovery improvement. Independent interpretation review remains pending; full results are in the companion evidence.

Changes from the proposal. We completed two seeds rather than the proposed three. Paracetamol replication is deferred until aspirin recovery is feasible. A revised adequacy protocol followed the preserved small-subset failures. Responsibilities changed; the representation check adds an extension. Long-range interactions beyond 7 Å remain future work, not an explanation established by these results.

Three next actions. First, finish the MACE loader and check its predictions against saved validation outputs. Run the unchanged ten-start and six-reference controls. Second, use that evidence to assess a separately declared procedure with a per-atom movement cap. Compare the same checkpoints and starts; label reused-validation analysis as a post-hoc follow-up.

Third, measure closed-loop work and prediction time on identical hardware. Closed-loop work totals force along a path returning to its start and checks energy consistency. Declare any matched-wall-time training budget before new runs.

Team contributions and reproducibility. Shedrack completed data preparation, model/training integration, the first MACE reference and current experiments. Olufunke authored the recovery evaluator. Emmanuel Aaron owns remaining physical/cost analysis; Muhammed Idris owns MACE loading, prediction checks and independent reproduction.

Assignments are not completed contributions. Code/evidence: GitHub PR #10, stacked on PRs #9 and #8. Before submission, confirm contributions, staff access to this private repository, live course instructions and final team approval.

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